

For all quadratics of the form  $ax^2 + bx + c$ , show

that

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$ax^2 + bx + c = 0$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \rightarrow x^2 + \frac{b}{a}x = -\frac{c}{a}$$

$\forall x \in \mathbb{R}, \exists \gamma, \mu \in \mathbb{R}$  such that

$$x = \gamma + \mu$$

Then we have

$$a(\gamma + \mu)^2 + b(\gamma + \mu) + c = 0$$

$$a\gamma^2 + 2a\gamma\mu + a\mu^2 + b\gamma + b\mu + c = 0$$

$$a\gamma^2 + \gamma(2a\mu + b) + a\mu^2 + b\mu + c$$

Let  $2a\mu + b = 0$ , then  $\mu = -\frac{b}{2a}$  and we have

$$a\gamma^2 + \gamma(0) + a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c = 0$$

$$a\gamma^2 + \frac{ab^2}{4a^2} - \frac{b^2}{2a} + c = 0$$

$$\gamma^2 + \frac{b^2}{4a^2} - \frac{b^2}{2a^2} + \frac{c}{a} = 0$$

$$\gamma^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$

$$\gamma^2 = \frac{b^2}{4a^2} - \frac{4ac}{4a^2} = \frac{b^2 - 4ac}{4a^2}$$

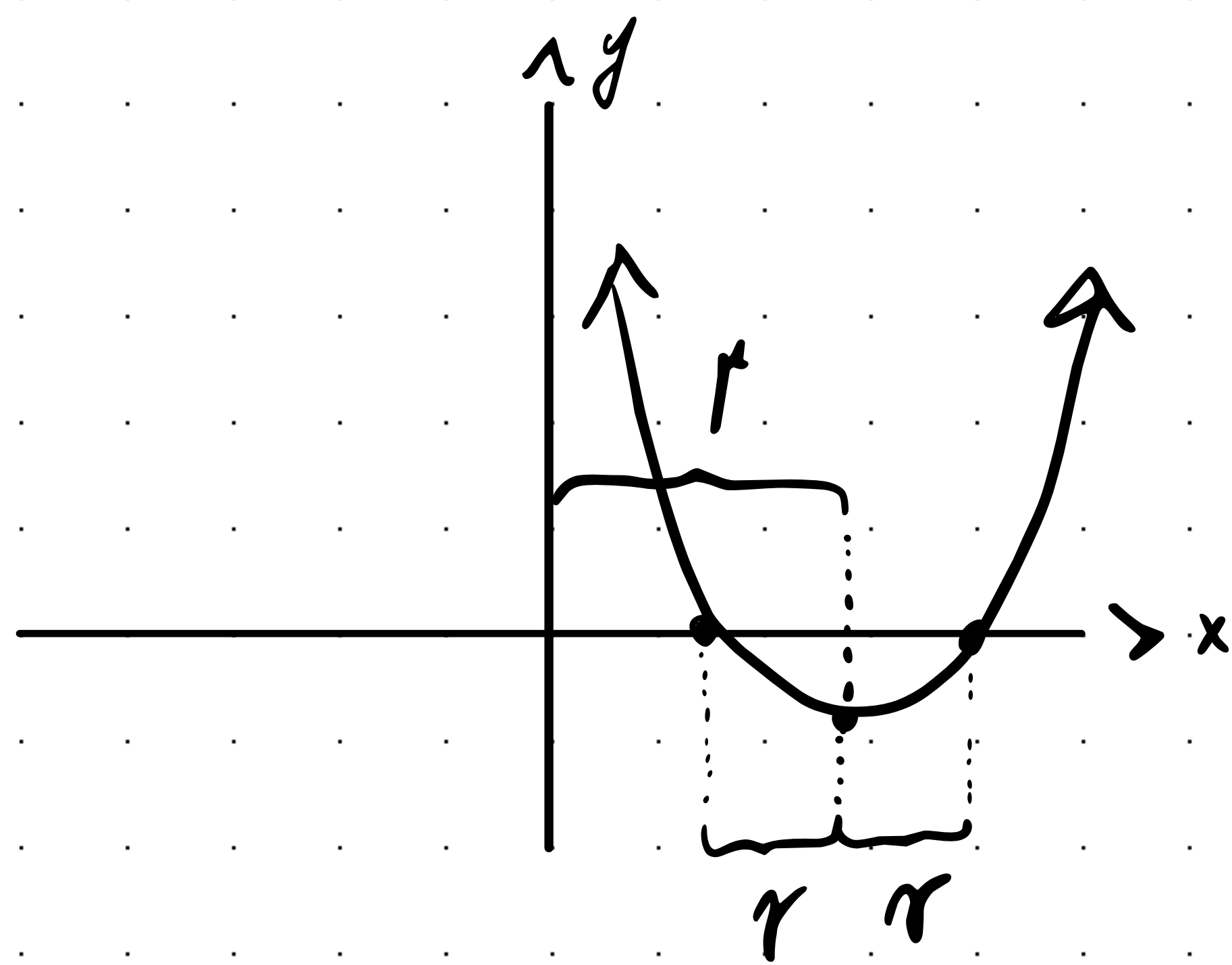
$$\gamma = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\gamma = x + \frac{b}{2a}, \text{ so } x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Q.E.D.

## Methodology

The linear ( $bx$ ) term of a quadratic prevents us from directly solving for  $x$ . To work around this, we shift the Parabola so that it is centered on the  $y$ -axis. This action eliminates the linear term. We then find where the shifted Parabola intersects the  $x$ -axis. We will call these values  $r$ . We then shift them by  $\mu$ , the horizontal distance the Parabola was shifted. This maps  $r$  back to the original  $x$ -intercept(s).



\*  $r$  is positive or negative, so we have

$$x = \mu + r$$

$$x = \mu + (-r) \\ = \mu - r$$

## Variable Analysis

$x$ :  $x$ -intercepts of the original Parabola

$r$ : distance of the  $x$ -intercept(s) from the vertex  $\mu$

$\mu$ : Horizontal distance the Parabola was shifted to become centered on the  $x$ -axis; the  $x$ -value of the Parabola's vertex.

$\mu^2 - \mu - c$ : adjusted constant that maps the Parabola over the  $y$ -axis while maintaining the same  $y$ -value of its vertex

## Alternate Approach - Equating Coefficients

Suppose  $r_1$  and  $r_2$  are solutions of  $ax^2 + bx + c = 0$ . Without the loss of generality, suppose that  $a > 0$  and  $r_1 \geq r_2$ , then

$$\begin{aligned} ax^2 + bx + c &= a(x - r_1)(x - r_2) \\ &= a(x^2 - (r_2 - r_1 + r_1)r_2) \\ &= ax^2 - a(r_1 + r_2)x + ar_1r_2 \end{aligned}$$

Inspect the form of the piece on the right side of the equation:  $ax^2 + bx + c$ . The Coefficients must be equal on both sides. We then have the following:

$$b = -a(r_1 + r_2) \rightarrow r_1 + r_2 = -\frac{b}{a}$$

$$c = ar_1r_2 \rightarrow \frac{c}{a} = r_1r_2$$

We also know that the Parabola's vertex is located at  $x = -\frac{b}{2a}$  and that its zeros are located an equal distance,  $r$ , from this point. We then have

$$r_1 = -\frac{b}{2a} + r$$

$$r_2 = -\frac{b}{2a} - r$$

We can then substitute these values into  $\frac{c}{a} = r_1 r_2$  to obtain

$$\frac{c}{a} = \left(-\frac{b}{2a} + r\right)\left(-\frac{b}{2a} - r\right) = \frac{b^2}{4a^2} - r^2$$

$$r^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}$$

$$r = \sqrt{\frac{b^2 - 4ac}{4a^2}} = \frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}} = \frac{\sqrt{b^2 - 4ac}}{2a}$$

We can now use this  $r$  value in terms of  $a, b, c$  to find  $r_1$  and  $r_2$  in terms of the same constants, as follows:

$$r_1 = -\frac{b}{2a} + r = -\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$r_2 = -\frac{b}{2a} - r = -\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a} = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Generalizing, } r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$